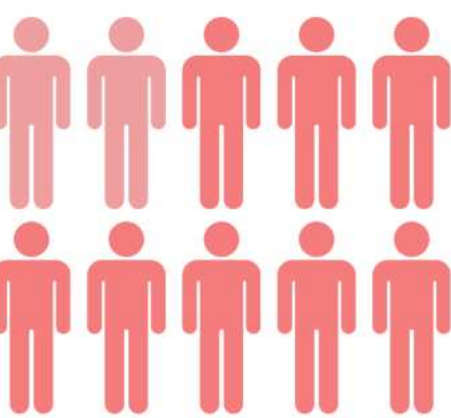


Investigating The Effects of Curcumin and *Ipomoea batatas* Root Extract on Cell Viability in Melanoma

BACKGROUND

Melanoma is a type of skin cancer that originates in melanocytes, the pigment-producing cells in our skin.



Melanoma accounts for **80%** of skin cancer deaths due to its metastatic, aggressive nature.

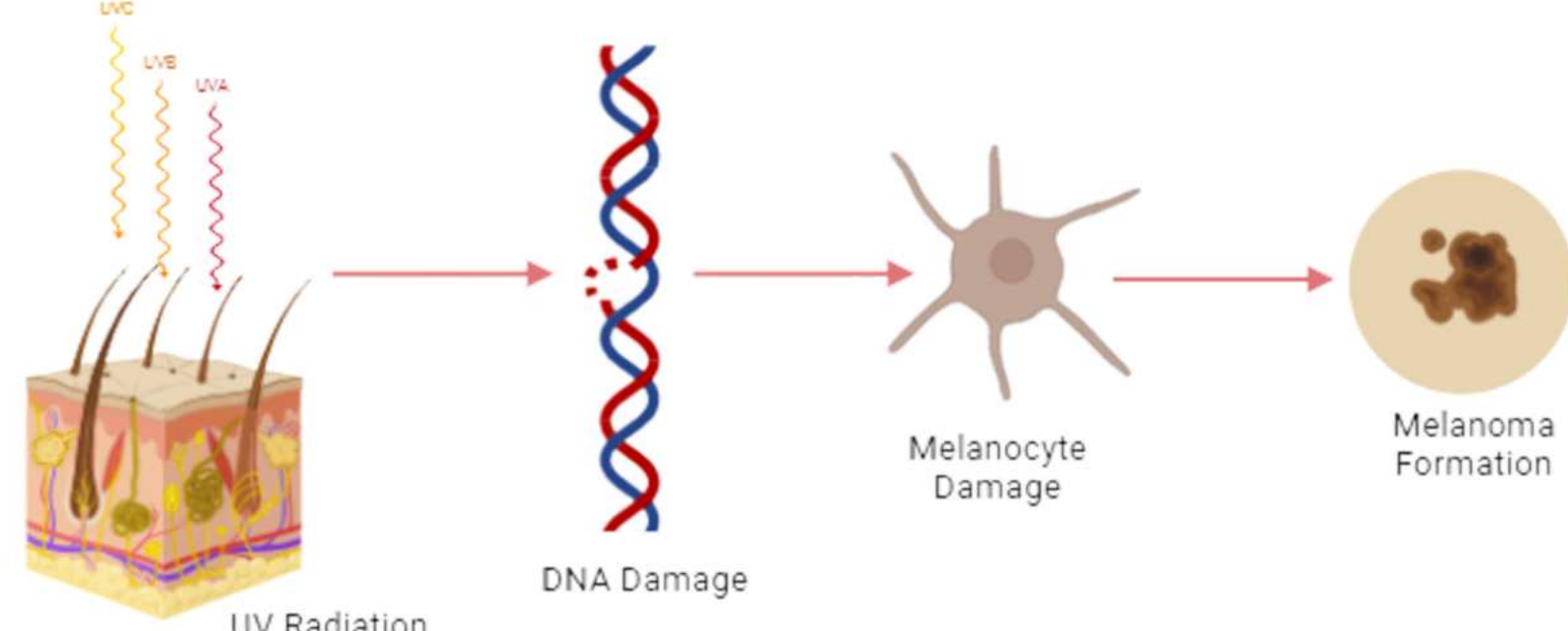


Figure 1. A visual representation of melanoma pathogenesis

Current melanoma treatments, including immunotherapy, chemotherapy, and surgery can be highly effective but often come with harsh side effects and high costs.



Curcumin is a **polyphenol** that has exhibited **antitumor properties** in lung cancer, glioblastoma, and melanoma.

Ipomoea batatas L. exhibits **anti-tumor** and **anti-cancer properties**, such as in prostate cancer cells.



PURPOSE

The purpose of this study is to investigate whether combining curcumin with *Ipomoea batatas* (IB) root extract would reduce melanoma cell viability more than individuals treatments.

HYPOTHESIS + VARIABLES

Combining curcumin and *Ipomoea batatas* root will reduce cell viability more than individual treatments. This reduction will occur in a dose-dependent manner, where higher concentrations lead to greater reductions.

IV: Concentrations of both treatments

DV: Cell viability, measured using acid phosphatase

Negative Control: Untreated (growth media)

Positive Control: Curcumin only

ACID PHOSPHATASE ASSAY

Cell viability will be measured using the acid phosphatase assay, which assesses cell metabolic activity by measuring absorbance as an indicator of viable cells.

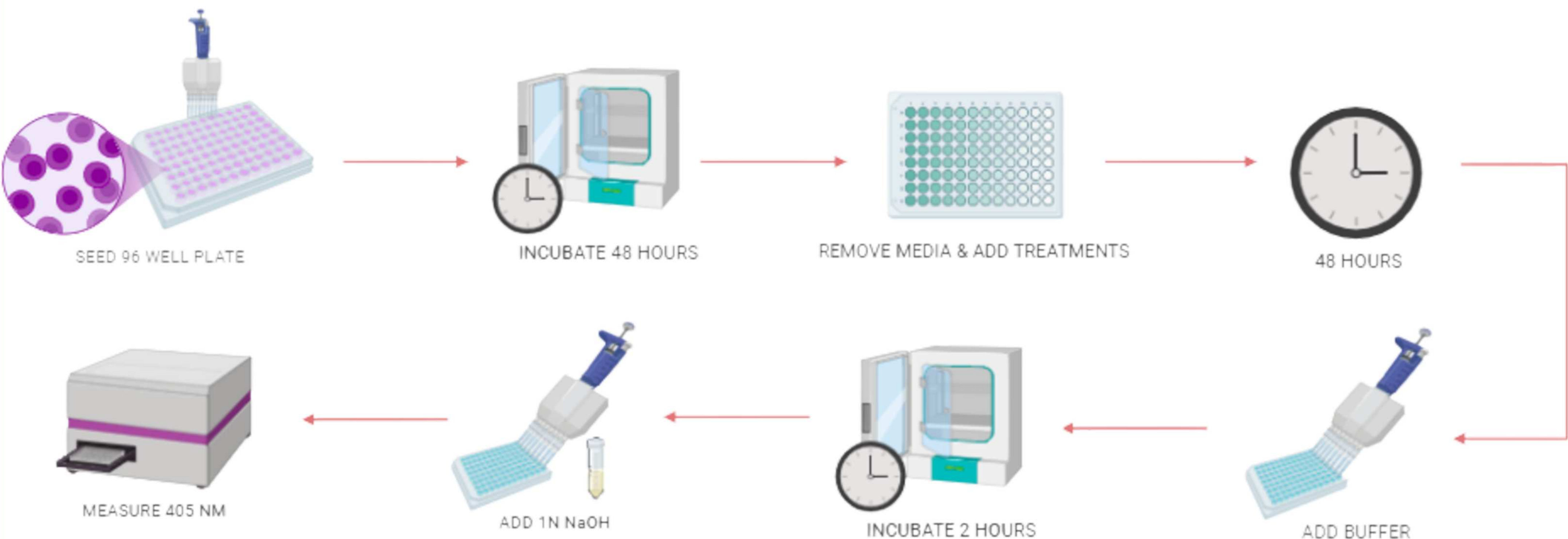


Figure 2. A visual representation of the acid phosphatase assay to measure cell viability.

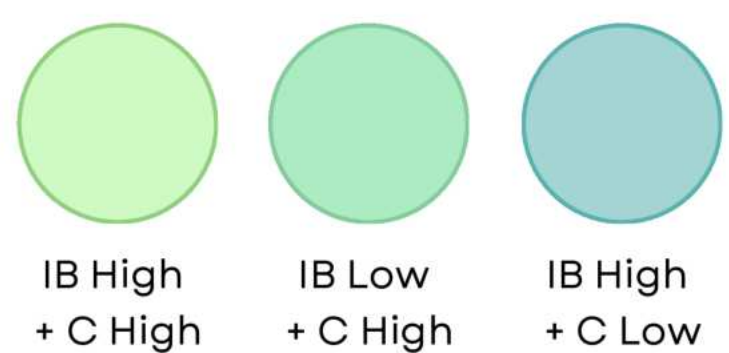
ASSAY BUFFER

- 0.1 M Na Acetate
- 0.1% Triton X-100
- 5 mM pNPP

CONCENTRATIONS



COMBINATIONS



RESULTS

ACID PHOSPHATASE RESULTS

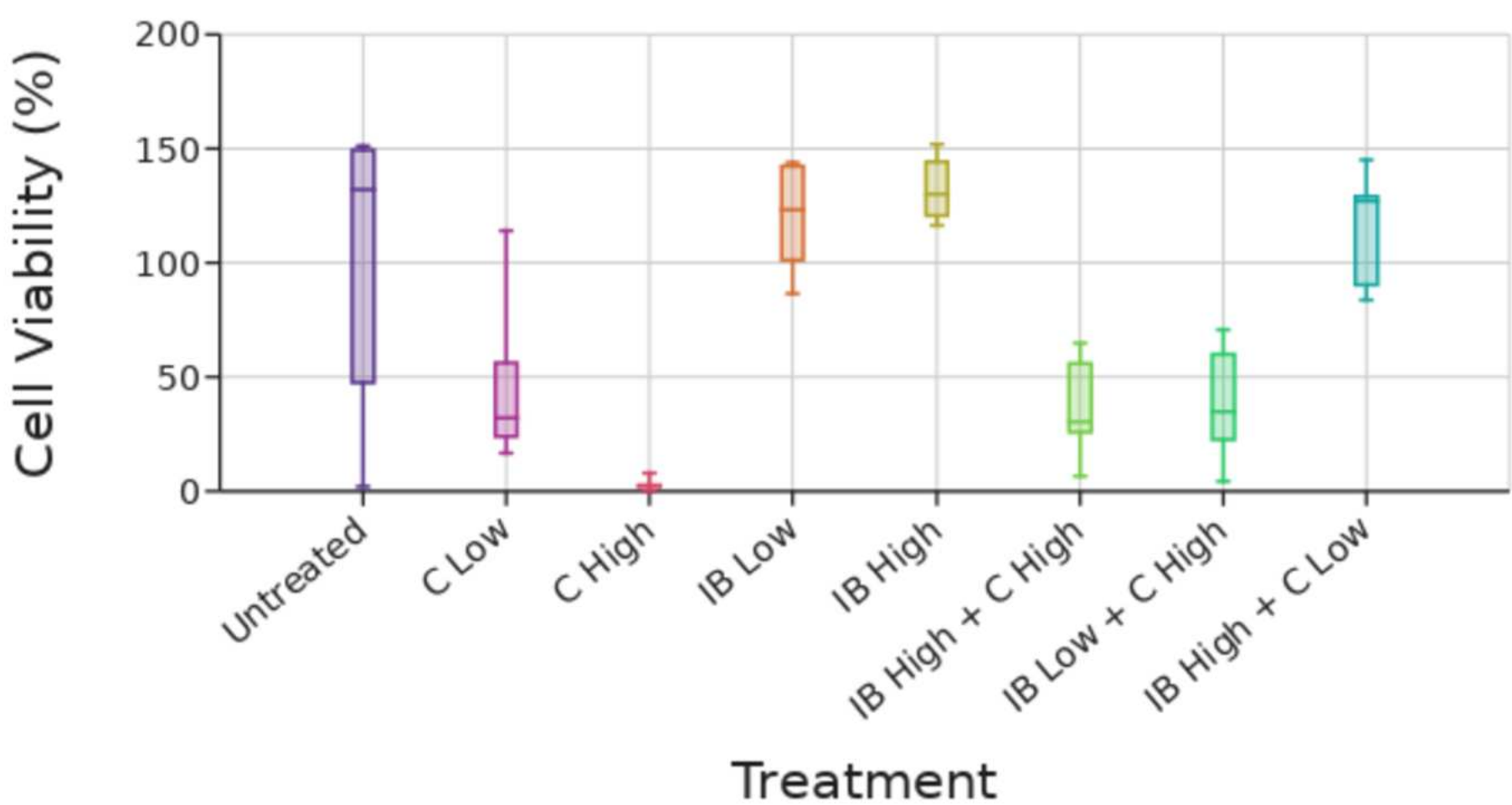


Figure 3. Box and whisker plot showing B16-F10 cell viability (%) across different treatment groups.

KRUSKAL-WALLIS TEST

Degrees of Freedom (df)	H-statistic	P-value	Interpretation of P
7	91.70	<0.01	A P-value of <0.01 means extremely strong confidence that the groups are different.

Table 1. Summary of Kruskal-Wallis test results comparing B16-F10 cell viability across treatment groups. This table includes the degrees of freedom (df), the H-statistic, p-value, and interpretation of P.

CONCLUSION

The Kruskal-Wallis test revealed a statistically significant difference between the treatment groups ($p < 0.01$), rejecting the null hypothesis that all treatments effect B16-F10 cell viability the same.

The Dunn's post hoc test and cell viability (%) data show that curcumin (25 and 100 ug/mL) significantly reduced cell viability compared to the negative control ($p < 0.01$). However, neither concentration of *Ipomoea batatas* (IB) reduced B16-F10 cell viability, and when combined with curcumin, the treatment did not further reduce viability ($p = 1.00$). In fact, the combination of curcumin and IB resulted in a higher median cell viability (%) values than curcumin alone.

These findings do not support the original hypothesis that combining curcumin and IB would work synergistically to reduce cell viability. However, because curcumin was highly effective at the tested concentrations, potential synergistic effects could not be observed.

FUTURE WORK

To properly evaluate synergy between the two treatments, future experiments should include lower concentrations of curcumin in combination with IB. During the first trial for testing the combined treatments, most wells produced overflow on the plate reader. As a result, only one full trial was used for analysis. In the future, repeating the experiment with more trials would improve the accuracy of the results. Additionally, exploring different time points (e.g. 72 hours) or using alternative assays could provide further insight into potential interactions.

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